

POINTE NOIRE, Congo

WMO: **644000**Lat: **4.816S**Lon: **11.887E**Elev: **55**StdP: **14.67**Time Zone: **1.00 (E01)**Period: **97-19**WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	66.3	67.9	63.9	89.0	72.3	64.4	90.8	72.4	14.5	76.7	13.9	76.8	1.6	100	0.293

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	11.3	89.5	79.7	88.2	79.6	87.8	79.6	81.4	85.7	80.8	85.5	80.3	85.0	9.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.4	158.1	84.3	79.2	151.9	83.4	79.0	151.0	83.3	45.1	86.4	44.6	86.5	44.1	85.5	
(p)															

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	12.8	11.8	64.0	91.1	1.5	1.1	62.9	91.9	62.0	92.5	61.2	93.1	60.1	93.9		
(5)			62.9	82.7	1.5	1.0	61.8	83.4	60.9	83.9	60.0	84.5	58.9	85.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.9	81.2	81.7	82.3	81.9	80.2	76.0	73.5	73.9	76.4	79.2	80.3	80.4
	DBStd	3.43	1.67	1.73	1.62	1.60	1.98	1.94	1.71	1.59	1.55	1.37	1.39	1.52
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10547	967	886	1002	956	938	780	727	742	793	906	909	941
	CDD65	5072	502	466	537	506	473	330	262	277	343	441	459	476
	CDH74	43169	5220	5097	6110	5443	4325	1628	684	708	1619	3556	4306	4473
	CDH80	10785	1538	1748	2270	1783	981	101	11	9	74	414	858	998
Wind	WSAvg	5.8	5.6	5.7	5.8	5.7	5.4	5.2	5.2	5.5	6.5	7.3	6.6	5.5
Precipitation	PrecAvg	48.7	6.9	8.3	8.0	5.7	2.0	0.0	0.0	0.1	0.6	2.9	8.1	6.0
	PrecMax	77.3	19.1	17.8	15.6	12.4	13.0	0.8	0.4	0.8	2.0	15.9	26.4	15.2
	PrecMin	18.0	0.4	0.4	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.3	0.2
	PrecStd	13.4	4.1	4.3	3.7	3.2	2.3	0.1	0.1	0.2	0.5	3.0	5.9	3.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.3	89.9	90.0	89.9	88.1	83.5	80.9	80.9	82.8	85.7	86.8	87.9
		MCWB	79.6	79.8	79.8	79.9	79.6	75.3	72.4	71.9	74.4	77.4	78.2	78.8
	2%	DB	88.0	88.9	89.5	88.6	86.4	82.4	79.4	79.3	82.1	84.3	86.0	86.3
		MCWB	79.5	79.8	79.7	79.7	79.0	74.8	71.7	71.3	74.1	76.8	78.0	78.5
	5%	DB	87.1	88.0	88.2	87.9	85.9	80.9	78.7	78.8	80.7	83.2	84.6	85.8
		MCWB	79.2	79.7	79.6	79.7	78.7	73.9	71.4	71.2	73.4	76.2	77.4	78.3
	10%	DB	86.0	87.4	87.7	86.6	84.4	80.3	77.3	77.3	79.2	82.4	84.2	84.4
		MCWB	78.7	79.5	79.6	79.3	78.1	73.6	70.7	70.8	73.2	75.8	77.3	77.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.1	81.9	82.3	81.9	80.9	77.3	74.2	73.6	76.3	79.0	79.8	80.1
		MCDB	85.7	85.8	86.1	85.8	84.8	80.4	77.7	77.3	79.8	82.5	83.5	84.1
	2%	WB	80.3	81.0	81.4	81.0	80.1	76.3	72.9	72.8	75.5	78.1	79.1	79.4
		MCDB	85.1	85.7	85.9	85.4	83.9	79.6	76.9	76.6	78.9	81.6	82.9	83.4
	5%	WB	79.8	80.5	80.7	80.5	79.5	75.6	72.5	72.3	74.8	77.4	78.7	79.0
		MCDB	84.6	85.6	85.7	85.1	83.4	79.0	76.4	76.1	78.3	81.0	82.6	83.1
	10%	WB	79.2	79.9	80.1	80.0	79.0	74.7	71.9	71.8	74.3	76.8	78.2	78.6
		MCDB	83.9	84.9	85.2	84.5	82.8	78.3	75.7	75.6	77.9	80.4	82.1	82.7
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.9	11.3	10.5	8.9	8.8	9.2	8.5	7.4	7.2	8.4	9.1
		MCDBR	11.1	11.9	11.8	11.6	9.8	9.3	10.2	9.4	8.3	8.1	9.1	10.3
		MCWBR	4.5	5.1	5.0	4.9	4.0	3.6	4.0	3.5	3.0	3.2	3.7	4.2
	5% WB	MCDBR	10.3	11.2	11.2	10.7	9.3	8.4	9.3	8.4	7.2	7.6	8.8	9.5
MCWBR		4.6	5.3	5.2	5.0	4.2	3.7	4.1	3.5	3.1	3.5	4.0	4.4	
Clear-Sky Solar Irradiance	taub	0.561	0.564	0.525	0.475	0.577	0.665	0.710	0.731	0.641	0.532	0.485	0.538	
	taud	2.031	2.030	2.155	2.319	1.963	1.737	1.640	1.617	1.807	2.120	2.273	2.098	
	Ebn at Noon	252	252	258	264	228	202	195	199	226	257	270	256	
	Edh at Noon	58	58	51	42	56	69	77	82	71	52	45	54	
All-Sky Solar Radiation	RadAvg	1487	1484	1615	1527	1303	1171	1143	1141	1113	1200	1398	1362	
	RadStd	76	98	87	79	104	85	71	91	52	55	87	102	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page